

MODULE 7A • EPISODE 1

Our Cosmic Neighborhood: Time, Seasons, and Celestial Observations

Clocks, the Equinox, the Harvest Moon, and the Winter Sky

TEACHER GUIDE

ASTRONOMY: OUR JOURNEY THROUGH THE COSMOS

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EPISODE OVERVIEW

Our Cosmic Neighborhood: Time, Seasons, and Celestial Observations

This episode starts close to home. It opens with the timekeeping astronomers depend on — the 24-hour clock, the Prime Meridian, and Coordinated Universal Time — then turns to the fall equinox and what it shows anyone with a clear horizon: equal day and night, a sunset due west, and a way to read your own latitude off the setting Sun. From there it moves to the Harvest Moon and the features visible on the full Moon, and closes with two star patterns of the autumn and winter sky, the Winter Triangle and Orion. Everything discussed can be checked with the unaided eye or a pair of binoculars.

The Panel

Speaker	Role
Dr. Michael Harrison	Host; astrophysicist
Dr. Alessandro Rossi	Cosmologist, European Space Agency
Dr. Amara Johnson	Theoretical physicist

Learning Objectives

After this episode you should be able to:

- ◆ Explain why astronomers use the 24-hour clock and record observations in UTC.
- ◆ Use the rule that 15° of longitude equals one hour of clock time to find a time difference.
- ◆ Describe what happens at the September equinox and why it is a single instant, not a day.
- ◆ Explain why an equinox day has slightly more than 12 hours of daylight, and define civil twilight.
- ◆ Explain why the equinox Sun sets due west for everyone, and how its slanted path gives your latitude.
- ◆ Explain the Harvest Moon, and identify the maria, highlands, Tycho, Copernicus, and the Sea of Tranquility.
- ◆ Distinguish an **asterism** from a **constellation**, using the Winter Triangle and Orion.
- ◆ Use star color as a rough thermometer, and explain why the Winter Triangle is a trick of perspective.

QUICK REFERENCE

Key Terms

Term	Definition
24-hour clock	Timekeeping that runs 0000 to 2359 hours, removing any AM/PM ambiguity. Astronomers use it exclusively.
Prime Meridian	The line of 0° longitude, running through Greenwich in London.
GMT / UTC	Greenwich Mean Time, the reference time at the Prime Meridian, and its more precise modern successor, Coordinated Universal Time.
Ecliptic	The Sun's apparent yearly path around the sky — really a reflection of Earth's orbit.
Equinox	The instant the Sun crosses Earth's Equator on the ecliptic. The word means “equal night.”
Zenith	The point directly overhead.
Terminator	The line dividing Earth's sunlit half from its dark half.
Twilight	Scattered sunlight before sunrise (dawn) and after sunset (dusk). Civil twilight is the brightest of its three stages.
Harvest Moon	The full Moon nearest the September equinox, which rises soon after sunset several evenings in a row.
Maria	The dark gray “seas” on the Moon — ancient lava plains. Singular: mare.
Moon illusion	The Moon looking larger near the horizon. It happens in the brain; the Moon's size does not change.
Asterism	A recognizable pattern of stars that is not an officially defined constellation.
Constellation	An officially defined region, or patch, of the sky.
Red supergiant	A massive star late in life that has swollen enormously and cooled at its surface.
White dwarf	The burned-out core left behind by a star like the Sun.
Celestial Equator	Earth's Equator projected onto the sky.
Stellar nursery	A cloud of gas and dust in which new stars are forming, such as the Orion Nebula.

QUICK REFERENCE

Numbers to Know

Value	What it refers to
24 h	Time for Earth to turn 360°
15°	Longitude equal to one hour of clock time
6 h	Usual difference between Chicago and London (5 h for a few weeks each spring and fall)
~12 h	Daylight on an equinox day — a few minutes more in practice
< 10 min	Extra equinox daylight at mid-latitudes
~30 min	Civil twilight at each end of the day at mid-latitudes
~13 h	Usable light on an equinox day at mid-latitudes, twilight included
~8.6 ly	Distance to Sirius
~11.5 ly	Distance to Procyon, the eighth brightest star
600+ ly	Distance to Betelgeuse, about 10 million years old

EPISODE CONTENT

Part-by-Part Summary

Part 1 — Keeping Time Like an Astronomer

An observation is only as good as its timestamp. Astronomers compare notes across continents, so they need one system that cannot be misread. Much of the world uses the **24-hour clock** professionally; astronomers use it exclusively. “1900 hours” can only mean evening.

They also need a common reference point on the globe: the **Prime Meridian**, 0° longitude, running through Greenwich in London. Its clock is the world's reference — **Greenwich Mean Time** and its more precise modern form, **Coordinated Universal Time (UTC)**. From there it is geography: Earth turns 360° in 24 hours, so 15° of longitude is one hour of clock time. Chicago, about 90° west of Greenwich, runs six hours behind it.

Daylight Saving Time complicates things in two ways. It moves local clocks an hour off standard time for part of the year, and countries do not all switch on the same date, so for a few weeks each spring and fall Chicago and London are five hours apart instead of six. Astronomers avoid the problem by recording everything in UTC and converting afterward.

Part 2 — The Fall Equinox

Over a year the Sun appears to travel around the sky along the **ecliptic**. It is really Earth orbiting the Sun, but the geocentric picture is convenient. Twice a year the Sun's path crosses Earth's Equator. In September it crosses from north to south, beginning astronomical fall in the Northern Hemisphere and spring in the Southern Hemisphere.

The **equinox** is a moment, not a day. It happens everywhere at the same instant, but the calendar date depends on where you live: a crossing an hour after midnight in London is still the previous evening in the Americas.

On that day the noon Sun stands at the **zenith** somewhere on the Equator. The **terminator** runs through both poles and cuts every circle of latitude in half, so nearly everyone gets about 12 hours of daylight and 12 of darkness — hence *equinox*, “equal night.”

In practice daylight runs a few minutes over 12 hours — under 10 minutes at mid-latitudes, more toward the poles. Sunrise and sunset are timed from the Sun's upper edge, not its center, and the atmosphere bends sunlight around Earth's curve so the Sun stays visible for a couple of minutes after it has geometrically set. Darkness does not follow at once either: **twilight** continues, and **civil twilight**, the brightest of its three stages, lasts about half an hour at each end of the day at mid-latitudes. Together that gives roughly 13 hours of usable light.

Part 3 — Due West and Your Latitude from the Equinox Sunset

On an equinox day the Sun rises due east and sets due west for essentially everyone on Earth, so the sunset shows you true west from your own backyard, no compass needed.

Why it works everywhere: point at the setting Sun and your arm aims not along your circle of latitude but along a **great circle** toward the place where the Sun is overhead at that instant — where it is local noon. On the equinox that spot is on the Equator, a quarter of the way around the world to your west: near Howland Island in the Pacific for Chicago, over the Galapagos Islands for London. Different spots, same direction.

The same sunset gives your latitude. Except at the Equator, the Sun does not drop straight down; it slides toward the horizon at a slant. The angle between that path and a vertical line rising from the point where it meets the horizon equals your latitude — 0° of lean at the Equator, 40° at 40° north.

Part 4 — The Harvest Moon and What to See on It

The full Moon nearest the September equinox is the **Harvest Moon**. Around then the full Moon rises soon after sunset several evenings in a row, which gave farmers extra evening light to bring in the crops.

To the unaided eye the dark gray patches are the **maria**, or “seas” — ancient lava plains — and the brighter areas are cratered highlands. A low Moon looks hazy and large for two different reasons. Near the horizon you look through far more atmosphere, which softens and reddens the view. And the large size is the **Moon illusion**, which happens in the brain; an hour later, higher up, the same Moon looks smaller and sharper.

Many telescopes turn the image upside down and some also swap left and right, while ordinary binoculars keep it upright and correct, so know your instrument before deciding a crater has moved. Landmarks: the crater **Tycho** in the bright southern highlands, with pale rays at full Moon; **Copernicus**, toward the center on the edge of the Ocean of Storms; and the **Sea of Tranquility**, the dark oval right of center, where Apollo 11 landed. The landing hardware itself is far too small for any telescope on or orbiting Earth to resolve.

Part 5 — The Winter Triangle

Three brilliant stars — **Betelgeuse**, **Sirius**, and **Procyon** — form a big, nearly equilateral triangle. It is an **asterism**, a recognizable star pattern, not an officially defined constellation. It rules the winter evening sky but rises before dawn in early autumn, so early risers get a preview months ahead.

Betelgeuse, the reddish star at Orion's shoulder, is over 600 light-years away and only about 10 million years old. It was born huge and blue-hot; massive stars burn through their fuel fast, and it has already swollen into a **red supergiant** that will end as a **supernova**.

Sirius, the Dog Star in Canis Major, is the brightest star in the night sky, partly because it is genuinely luminous and partly because it is close, about 8.6 light-years. It has a **white dwarf** companion, the burned-out core of what was once the more massive star of the pair. Procyon, in Canis Minor, is the eighth brightest star, about 11.5 light-years away, and has a white dwarf companion too.

The triangle is a trick of perspective. At roughly 8, 11, and 600-plus light-years, the three stars have nothing to do with one another; they line up only because of where we happen to be.

Part 6 — Orion

Find Orion by its **belt**: three bright, evenly spaced stars in a short straight row, called the Three Kings in some traditions. From the belt the figure assembles itself, with ruddy Betelgeuse at one shoulder and blue-white **Rigel** at the diagonally opposite foot.

Orion is two things at once: a **constellation**, an official patch of sky, and an **asterism**, the hunter figure traced inside it. Because the belt sits almost exactly on the **Celestial Equator**, Orion is visible from nearly everywhere on Earth — he simply stands on his head for observers in the Southern Hemisphere.

Below the belt hangs the **sword**, a short vertical line of fainter stars that can be hard to see from a city, though a phone camera's night setting usually picks them up. The middle “star” is the **Orion Nebula**, a huge cloud of gas and dust where new stars are forming — one of the closest stellar nurseries. Binoculars show a soft glow; a telescope begins to show structure.

Compare colors while you are there: Betelgeuse is orange-red, Rigel and the belt stars blue-white. Star color works as a thermometer — reddish stars are cooler, blue-white stars far hotter.

STUDY QUESTIONS

Questions with Model Answers

Question numbering matches the student study guide exactly. Model answers are written at the level expected of a strong student response, not as the only acceptable wording.

Red numbers mark core questions — the ones to be sure you can answer.

Part 1 — Keeping Time Like an Astronomer

- 1.** Why do astronomers use the 24-hour clock exclusively?

ANSWER It removes AM/PM ambiguity. “1900 hours” can only mean 7 p.m., so observations shared across countries cannot be misread.

- 2.** What is the Prime Meridian, and how are GMT and UTC related to it?

ANSWER The Prime Meridian is 0° longitude, running through Greenwich in London. GMT is the reference time there; UTC is its more precise modern successor and is the world's time standard.

- 3.** Why does 15° of longitude correspond to one hour of clock time?

ANSWER Earth turns 360° in 24 hours, and $360^\circ \div 24 \text{ h} = 15^\circ$ per hour.

- 4.** A city lies 75° west of Greenwich. Ignoring Daylight Saving Time, how many hours behind UTC is its standard time?

ANSWER $75^\circ \div 15^\circ \text{ per hour} = 5$ hours behind UTC. (New York's Eastern Standard Time is an example.)

- 5.** Give the two ways Daylight Saving Time complicates timekeeping, and say how astronomers avoid the problem.

ANSWER It shifts clocks an hour away from standard time for part of the year, and countries switch on different dates, so the usual gap between two cities (Chicago and London: six hours) is briefly different (five hours). Astronomers record in UTC and convert afterward.

Part 2 — The Fall Equinox

1. What is the ecliptic, and what is really happening when the Sun appears to move along it?

ANSWER The ecliptic is the Sun's apparent yearly path around the sky. The motion is really Earth orbiting the Sun; describing it from Earth's point of view is just convenient.

2. The equinox is described as “a moment, not a day.” Explain, and explain why its calendar date can differ from place to place.

ANSWER The equinox is the single instant the Sun crosses the Equator, and it happens everywhere at once. Local clocks differ, though, so if the instant falls just after midnight in London, it is still the previous evening in the Americas — a different calendar date.

3. What does the word *equinox* mean, and what geometry on Earth makes it true?

ANSWER “Equal night.” The terminator runs through both poles and cuts every circle of latitude exactly in half, so nearly every place gets about 12 hours of daylight and 12 of darkness.

4. Give the two reasons daylight on an equinox day runs a few minutes more than 12 hours.

ANSWER Sunrise and sunset are timed from the Sun's upper edge, not its center; and the atmosphere bends (refracts) sunlight around Earth's curve, so the Sun remains visible a couple of minutes after it has geometrically set.

5. What is civil twilight, and why does a mid-latitude equinox day give about 13 hours of usable light?

ANSWER Twilight is sunlight scattered down by the atmosphere before sunrise and after sunset; civil twilight is its brightest stage. At mid-latitudes it lasts about 30 minutes at each end, and two half-hours added to about 12 hours of daylight gives roughly 13 hours.

Part 3 — Due West and Your Latitude from the Equinox Sunset

1. Where does the Sun rise and set on an equinox day, and how can you use that?

ANSWER It rises due east and sets due west for essentially everyone on Earth, so an equinox sunset shows you true west from your own location without a compass.

2. When you point at the setting equinox Sun, toward what place on Earth is your arm aimed?

ANSWER Along a great circle toward the spot where the Sun is overhead at that instant — where it is local noon. On the equinox that spot is on the Equator, a quarter of the way around the world to the west.

3. The episode names the “local noon” spot for Chicago and for London. What are they, and why is the direction due west in both cases?

ANSWER Near Howland Island in the Pacific for Chicago; over the Galapagos Islands for London. Each is on the Equator 90° of longitude to the west, and the great circle from any latitude to that point leaves the observer heading due west.

4. How can the setting Sun's path tell you your latitude? What would you see at the Equator and at 40° north?

ANSWER The Sun slides down at a slant; the angle between its path and the vertical at the point where it meets the horizon equals your latitude. At the Equator it drops straight down (0° lean); at 40° north it comes in with a 40° lean.

Part 4 — The Harvest Moon and What to See on It

1. What is the Harvest Moon, and how did it get its name?

ANSWER The full Moon nearest the September equinox. Around then the full Moon rises soon after sunset several evenings in a row, giving farmers extra evening light to bring in the crops.

2. What are the dark and bright regions seen on the full Moon with the unaided eye?

ANSWER The dark gray areas are the maria, or “seas” — ancient lava plains. The brighter areas are cratered highlands.

3. A low Moon looks both hazy and unusually large. Give the separate cause of each.

ANSWER The haze and reddening come from looking through far more atmosphere near the horizon. The large size is the Moon illusion, which happens in the brain — the Moon's actual size does not change, and it looks smaller and sharper an hour later when higher.

4. Why should you check your instrument before comparing a telescope view with a Moon map?

ANSWER Many telescopes invert the image and some also reverse left and right, while ordinary binoculars keep it upright and correct. A feature may seem to be in the wrong place only because of the optics.

5. Name and locate the three lunar landmarks from the episode. Why can't a telescope show the Apollo 11 landing hardware?

ANSWER Tycho, in the bright southern highlands, with pale rays at full Moon; Copernicus, toward the center on the edge of the Ocean of Storms; and the Sea of Tranquility, the dark oval right of center, where Apollo 11 landed. The hardware is far too small at the Moon's distance for any telescope on or orbiting Earth to resolve.

Part 5 — The Winter Triangle

1. Which three stars form the Winter Triangle, and why is it called an asterism rather than a constellation?

ANSWER Betelgeuse, Sirius, and Procyon. It is a recognizable pattern of stars, not one of the officially defined regions of sky.

2. The Winter Triangle is a winter evening sight. When can you see it in early autumn?

ANSWER Before dawn. Early risers can see it months before it appears in the evening sky.

3. Betelgeuse is only about 10 million years old, yet it is near the end of its life. Explain.

ANSWER It was born very massive, huge and blue-hot, and massive stars burn through their fuel fast. It has already swollen into a red supergiant and will end as a supernova.

4. Give two reasons Sirius is the brightest star in the night sky, and describe its companion.

ANSWER It is genuinely luminous, and it is close, about 8.6 light-years. Its companion is a white dwarf, the burned-out core of the star that was once the more massive of the pair.

5. Why is the Winter Triangle called “a trick of perspective”?

ANSWER Its stars lie at about 8.6, 11.5, and 600-plus light-years. They are unrelated and only appear to form a triangle because of where we happen to view them from.

Part 6 — Orion

1. How do you find Orion, and where are Betelgeuse and Rigel in the figure?

ANSWER Find the belt: three bright, evenly spaced stars in a short straight row. Betelgeuse, orange-red, is at one shoulder; blue-white Rigel is at the diagonally opposite foot.

2. In what sense is Orion both a constellation and an asterism?

ANSWER As a constellation it is an officially defined patch of sky; as an asterism it is the hunter figure traced by its bright stars inside that patch.

3. Why can Orion be seen from nearly everywhere on Earth, and how does it look from the Southern Hemisphere?

ANSWER Its belt lies almost exactly on the Celestial Equator, which is visible from nearly every latitude. From the Southern Hemisphere the figure appears upside down.

4. What is the middle “star” of Orion's sword, and what do binoculars and a telescope show?

ANSWER The Orion Nebula, a vast cloud of gas and dust where new stars are forming — one of the closest stellar nurseries. Binoculars show a soft glowing patch; a telescope begins to show its structure.

5. What does star color tell you? Use Betelgeuse and Rigel as examples.

ANSWER Color indicates surface temperature: reddish stars are cooler, blue-white stars far hotter. Orange-red Betelgeuse is cool; blue-white Rigel is much hotter.

TEACHER NOTES

Corrections and Cautions

Points in the recorded episode that need a correction, a qualification, or a prepared answer for the question a sharp student will ask.

C1 “Seeing the lunar carriage” — likely a script or caption slip.

The Apollo 11 line probably intended *lunar lander* or *lunar module*; the descent stage is what remains at Tranquility Base. The point stands either way: the hardware is a few meters across, far below what any Earth-based or Earth-orbiting telescope can resolve. Worth noting that a spacecraft in lunar orbit (NASA's Lunar Reconnaissance Orbiter) *has* photographed the site — the episode's wording correctly limits the claim to telescopes on or around Earth.

C2 “Daylight Savings Time” — the official name is Daylight Saving Time.

Singular *saving*. Common error; worth correcting in the captions.

C3 Betelgeuse's “left shoulder.”

Traditionally Betelgeuse (Alpha Orionis) marks Orion's *right* shoulder, because the figure is drawn facing us. It sits on the upper *left* as a Northern Hemisphere viewer sees it. Students comparing with star charts may notice. The student guide says only “one shoulder.”

C4 Tycho's rays “to its west side.”

Tycho's ray system spreads across much of the Moon's near side, with its longest rays running northward across the maria. Students should expect rays in many directions rather than to one side. The student guide describes the rays without a direction.

C5 Betelgeuse's distance is uncertain.

“Over 600 light-years” reflects older estimates; a 2020 study put it near 550 light-years, and published values range from about 500 to 700. Present it as roughly 550–650 light-years if a student asks. The perspective point is unaffected.

C6 “Three Sisters” for Orion's belt.

The Three Kings is common; the more widely cited alternative is the Three Marys (Tres Marías) in Spanish- and Portuguese-speaking traditions. The student guide keeps only the Three Kings.

C7 Chicago and the 90° meridian.

Chicago is at about 87.6° W, not exactly 90°. The six-hour offset comes from the Central time zone, which is based on the 90° W meridian. Useful if a student computes $87.6 \div 15$ and gets 5.84 hours.

C8 “A couple of weeks each spring and fall.”

The US switches about three weeks before Europe in March and about one week after it in October/November, so the five-hour window is about three weeks in spring and one week in fall.

C9 Dr. Johnson's specialty differs between the intro card and the host's introduction.

The opening card calls her a theoretical physicist; the host adds “specializing in cosmology.” The series notes describe her as specializing in observational astronomy. The guides list her simply as a theoretical physicist.

TEACHER NOTES**Teaching the Episode****Make time zones a calculation, not a fact.**

Give students three longitudes and have them compute the offset from UTC by dividing by 15, then compare with actual time zones. The mismatch (Chicago at 87.6° W) opens a short discussion of why zones follow political boundaries.

Assign an equinox sunset.

If the timing works, have students note where the Sun sets from home on or near an equinox and mark that direction as west. The latitude-from-slant idea can be checked roughly with a protractor held at arm's length, or from a phone photo taken as the Sun nears the horizon.

Use a globe and a flashlight for the terminator.

Hold the light level with the equator and rotate the globe: every latitude circle is cut in half. Tilt the light to show solstice conditions for contrast.

Test the Moon illusion directly.

A coin or pill held at arm's length covers the Moon equally well near the horizon and high in the sky. Students who try it usually stop arguing about whether the illusion is real.

Contrast asterism and constellation with one diagram.

Draw Orion's official boundary and the stick figure inside it, then place the Winter Triangle across three constellations. The difference becomes obvious.

Preview Episode 7AE2.

The next episode visits Venus and Mars, the Moon's origin and phases, and the Milky Way and beyond. The Harvest Moon discussion here is a natural bridge to phases.